
TML26 - WATERMARK FORGERY ATTACK

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1 INTRODUCTION

Watermark forgery steals a watermark signal from genuine watermarked content and transplants it onto unrelated, malicious content so it is falsely detected as authentic. We operate in an almost black-box setting: for each of 8 unidentified watermarking methods we are given 25 images sharing one embedded message, with no knowledge of the embedding algorithm, key, or message. We must forge this watermark onto 200 clean targets (a fixed batch of 25 per method) such that a hidden detector recovers the message with high bit accuracy (S_{det}) while the forged images stay visually close to the originals (LPIPS-based S_{qlt}). Since the score is the product $S_{det} \times S_{qlt}$, a viable attack must extract a genuinely transferable watermark signal rather than simply corrupting the image.

2 APPROACH

2.1 INITIAL APPROACH: DEEP IMAGE PRIOR EXTRACTION WITH JND MASKING

As a first pass, we implemented a more elaborate pipeline based on a Deep Image Prior (DIP): for each watermarked source image, a small untrained encoder-decoder CNN is fit via 1000 gradient steps to reconstruct it, exploiting untrained CNNs' bias toward learning low-frequency structure before high-frequency detail, so the residual (image – network output) approximates the watermark. Per-image residuals are averaged across each group's 25 images, as in our final method. Injection uses a Sobel-edge JND mask (less signal in smooth regions, more in textured ones) and a per-image guardrail that cuts injection strength by 20% (up to 10 tries) until LPIPS falls under a strict 0.015 threshold. This scored **0.40** on the leaderboard.

2.2 FINAL METHOD: AVERAGED-RESIDUAL WATERMARK ESTIMATION

We then developed a substantially simpler method that estimates the watermark for each group WM_i in isolation from content, using a fixed classical denoiser and averaging residuals across the 25 images in each group:

$$\text{residual}_j = \text{watermarked}_j - \text{denoise}(\text{watermarked}_j), \quad \hat{w}_i = \frac{1}{25} \sum_{j=1}^{25} \text{residual}_j$$

Averaging over 25 images with different content but the same watermark cancels the image-specific component while reinforcing the shared watermark component. Targets are forged as

$$\text{forged} = \text{clip}(\text{target} + \alpha \cdot \hat{w}_i, 0, 255)$$

Why the simpler method outperformed the more elaborate one. Both approaches share the same structure: extract a per-image residual, average over 25 images per group, inject into targets. The DIP pipeline is strictly more sophisticated, a learned per-image denoiser and spatially-adaptive, guardrailed injection, yet scored only 0.40 while our simpler method reached 0.4260. We attribute this to the guardrail: with a strict per-image LPIPS threshold of 0.015 and a 20%-per-attempt cutback policy, the effective alpha applied is likely pulled well below its configured value of 5, since $S_{qlt} = e^{-8 \cdot \text{LPIPS}}$ already scores 0.85–0.90 at LPIPS around 0.02, only marginally above the cutoff, so the guardrail trades away real detection strength for a quality margin with diminishing returns. A 1000-step DIP fit may also converge less stably than a well-tuned classical denoiser, and is far costlier (200 independent CNN optimizations vs. one filter call). This suggests careful calibration of a simple method can dominate architectural sophistication here.

Hyperparameters. We used Non-Local Means (NLM) denoising, since it estimates an edge-aware “clean” image rather than uniformly blurring content, reducing leakage of real image detail into the residual. We fixed a single global strength $\alpha = 1.5$ across all 8 groups, chosen from an LPIPS sweep ($\alpha \in \{0.5, 1, 1.5, 2, 2.5, 3\}$) as a value that stayed visually safe across most groups while remaining simple enough to generalize.

2.3 EXPLORATORY EXPERIMENTS

Three diagnostics beyond the final method (full code in the repository). **(1)** Averaging raw images before denoising, instead of denoising then averaging, produced higher-variance patterns; a negative control on 25 *unwatermarked* images showed comparable variance, revealing this mostly captured a generic averaging artifact, so we discarded it. **(2)** The same control test on our final method showed only 4 of 8 groups (WM_1, 3, 5, 6) carry residual signal clearly above baseline under NLM, while WM_2 and WM_7 showed none under any denoiser tested; per-group tuning built on this did not outperform the simpler global setting, so we kept the latter. **(3)** A per-block DCT variance analysis flagged 40/192 coefficients in WM_3 (a coherent mid-frequency cluster, suggestive of coefficient quantization) versus at most 1 elsewhere; a targeted DCT forge for WM_3 achieved near-perfect quality (S_{qtl} proxy = 0.985) but did not move the aggregate score.

3 RESULTS

Method	Key detail	Score
DIP + JND masking (initial approach)	learned denoiser, guardrailed α	0.40
Averaged-residual (final method)	NLM, fixed $\alpha = 1.5$	0.4260 (best)
Averaged-residual (rerun)	identical code	0.4230
Per-group tuning / DCT-only	diagnostics above	≤ 0.4260

Moving from the DIP-based pipeline to the simpler fixed-denoiser method gave the largest single improvement ($0.40 \rightarrow 0.4260$, $\sim 6.5\%$ relative) despite the DIP method being architecturally more sophisticated (Section 2.3). The leaderboard keeps only a team’s best score; a later, code-identical rerun of our method scored 0.4230, a small difference we attribute to server-side evaluation noise rather than any code change. Subsequent tuning of our method stayed within this same band, suggesting our black-box residual approach, using only 25 samples per group, reaches a ceiling that further tuning does not exceed; at least 4 of 8 schemes carry extractable signal while at least 2 appear largely resistant to the pixel- and frequency-domain attacks we tried.

4 CONCLUSION

Having forged watermarks under black-box conditions for a subset of unidentified schemes using only 25 reference images each, we consider this a genuine, practically relevant vulnerability. Any system using watermarking as a trust signal, for provenance tracking, deepfake attribution, or proving content is AI-generated, implicitly assumes a small sample of watermarked outputs cannot be used to transplant that watermark onto unrelated content. Our results show this does not hold uniformly: several schemes leaked a transferable pattern from as little as 25 samples using only generic denoising and averaging, with no knowledge of the underlying algorithm. An attacker able to scrape a modest number of watermarked outputs from a public system, without ever touching its weights, keys, or detector, may be able to forge convincing fake content that passes as authentic. Conversely, schemes that resisted every method we tried (WM_2, WM_7) suggest robustness is achievable and correlates with how strongly an embedding is coupled to individual image content rather than reducible to a stable, low-dimensional signal across images. This argues for content-adaptive watermark design, and for treating robustness against averaging-based collusion attacks as a first-class evaluation criterion for any deployed watermarking system.

Code: GitHub Repository.

REFERENCES

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